FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IX
PAPER –1
TEST DATE: 17-08-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A *(01 – 04, 20 – 23, 39 – 42)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 24 – 29, 43 – 48)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 13, 30 – 32, 49 – 51)*: This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C *(14 – 19, 33 – 38, 52 – 57)*: This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A circular disc of radius R caries surface charge density $\sigma(r) = \sigma_0\left(1-\frac{r^2}{R^2}\right)$, where σ_0 is a constant and r is the distance from the centre of the disc. Assume that disc is in xz-plane and its axis is y-axis. We have an imaginary hollow sphere of radius R and centre at $\left(0, \frac{3}{5}R, 0\right)$. What is the flux of electric field through this imaginary sphere?

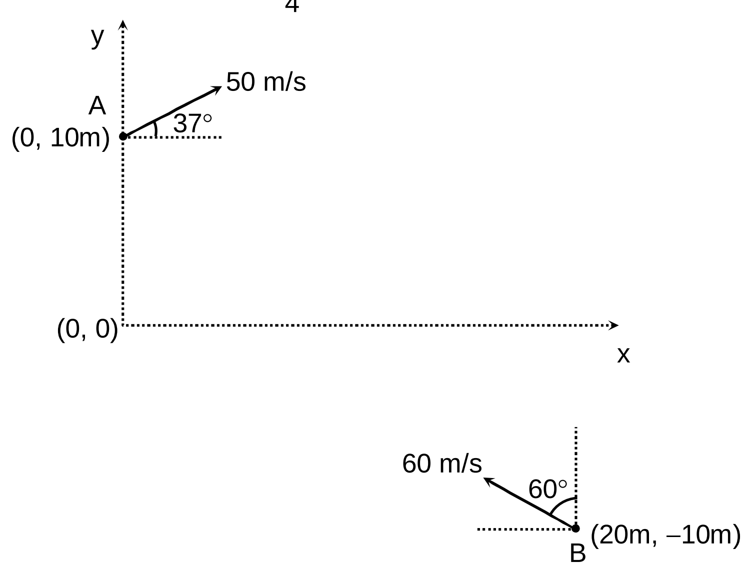
(A) $\frac{408}{625}\left(\frac{\sigma_0\pi R^2}{\varepsilon_0}\right)$

(B) $\frac{272}{625}\left(\frac{\sigma_0\pi R^2}{\varepsilon_0}\right)$

(C) $\frac{544}{625}\left(\frac{\sigma_0\pi R^2}{\varepsilon_0}\right)$

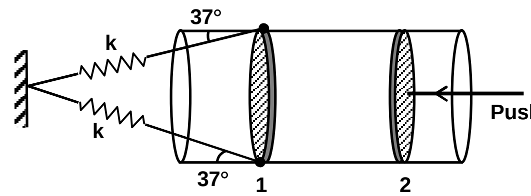
(D) $\frac{136}{625}\left(\frac{\sigma_0\pi R^2}{\varepsilon_0}\right)$

2. Two particles A and B are projected in the vertical plane with velocity 50 m/s at an angle of 37° with horizontal and with velocity 60 m/s at an angle of 60° with vertical respectively as shown in the figure. Find the minimum distance between the particles A and B during the motion? (Take g = 10 m/s^2 and $\tan 37° = \frac{3}{4}$)



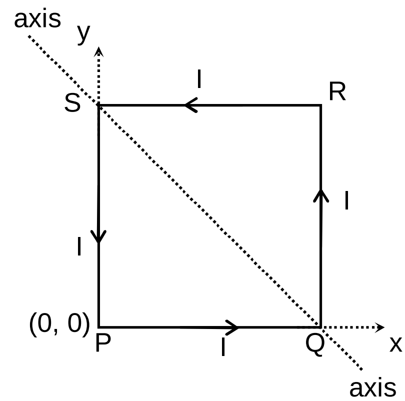
(A) 10 m
(B) 15 m
(C) 20 m
(D) 8 m

3. A long container has air enclosed inside at room temperature and atmospheric pressure 10^5 N/m^2 between two movable pistons. It has a volume 20,000 cc. The area of cross section is 100 cm^2 and force constant of springs is k = 781.25 N/m^2. Now push the right piston isothermally and slowly till it reaches the original position of the left piston-1 which is movable. What is the final length of air column. Assume that spring is initially relaxed and length of spring is very large as compare to separation between two piston.



(A) 100 cm
(B) 75 cm
(C) 50 cm
(D) 200 cm

4. A uniform rigid, square loop of mass 'm' and side 'ℓ' is free to rotate about an axis in xy plane passing through two corner as shown in figure. A constant current I is flowing in the loop. In space magnetic field $\vec{B}(x) = B_0\left(1+\frac{x}{\ell}\right)(\hat{i}+\hat{k})$ is present. Initially loop is held at rest in xy-plane. On releasing its instantaneous angular acceleration will be (space is gravity free space)



(A) $\frac{9IB}{\sqrt{2}m}$
(B) $\frac{5IB}{2\sqrt{2}m}$
(C) $\frac{9IB}{2\sqrt{2}m}$
(D) $\frac{5\sqrt{2}IB}{2m}$

Section – A (Maximum Marks: 24)

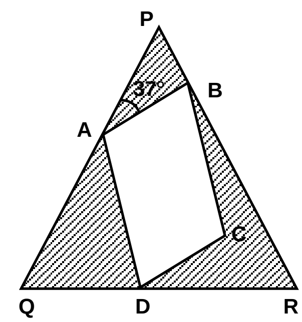
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A gaseous mixture at 300 K and 2×10^5 N/m^2 pressure contains 6g of H_2 and 8g of He. The mixture is expanded four times its initial volume, through an isobaric heating process. Then it is **isochoricaly** cooled unit its temperature again becomes 300 K. After that the gas mixture is isothermally compressed to its original volume. Choose the correct option(s). (take ℓn2 = 0.695)

(A) Ratio of molar specific heat (γ) of the mixture is $\frac{31}{21}$
(B) Ratio of heat absorb in isobaric expansion to that of heat rejected in isochoric cooling is $\frac{31}{21}$
(C) Heat rejected in isothermal compression is more than the heat rejected in isochoric cooling.
(D) Efficiency of the complete cycle is $\frac{161}{930}$.

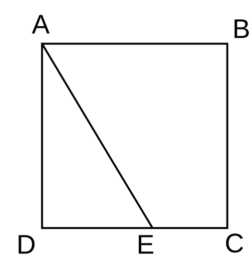
6. An electron, initially at rest, is released far away from a proton (fixed in space). The de-broglie wavelength of the electron when it is at distance r from proton is λ. The ratio of its Kinetic energy at this distance with the kinetic energy of the electron in ground state of hydrogen atom (Bohr model) with the radius of its orbit being r, is μ. Pick the correct option(s).
(A) $\lambda \propto \sqrt{r}$
(B) $\lambda \propto \frac{1}{r}$
(C) $\mu = \frac{1}{2}$
(D) $\mu = 2$

7. A rectangular cavity ABCD is carved inside an equilateral prism PQR of refractive index $\left(\frac{8}{5}\right)$ as shown in figure. If δ_{min} is the deviation in a ray entering at face PQ and emerging at face PR, without passing through AB or CD, then which of the following is/are correct.



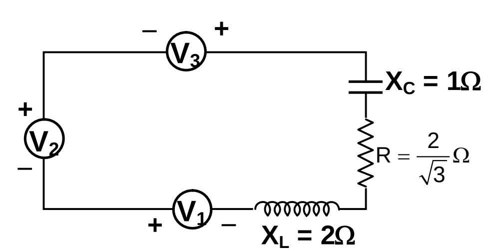
(A) There is no effect of cavity of the value of δ_{min}.
(B) The minimum deviation $\delta_{min} = 46°$
(C) If the cavity is filled with water ($\mu_W = 4/3$), the δ_{min}=46°
(D) If the cavity is filled with a liquid of refractive index $\mu = 2$, then $\delta_{min} = 23°$

8. ABCD is square as shown in figure, resistance of side AB, BC, CD and DA are 8Ω, 2Ω, 14Ω and 22Ω respectively. A point E lies on the CD such that a uniform wire of resistance 32Ω is connected across AE and constant potential difference is applied across A and C then B and E are equipotential. Choose the correct option(s).



(A) Resistance of DE is 10Ω
(B) Resistance of EC is 10Ω
(C) Equivalent resistance across AC is $\frac{20}{3}\Omega$
(D) Equivalent resistance across AC is 10Ω

9. Three alternating voltage sources $V_1 = 3\sin\omega t$, $V_2 = 5\sin(\omega t + 30°)$ and $V_3 = 5\sin(\omega t - 127°)$ volt connected across a resistor, inductor and capacitor as shown in the figure, then choose the correct option(s)



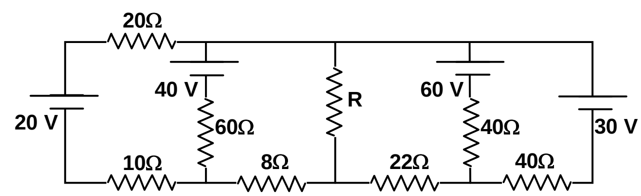
(A) The maximum current in the resistors is 3 ampere.
(B) The maximum current in the resistor is 2 ampere.
(C) Effective impedance of the circuit is $\sqrt{\frac{7}{3}}\Omega$
(D) Maximum potential of the three AC source is $\sqrt{21}$ volts.

10. Two vernier calipers A and B both have M.S.D. of 1 mm. 10 vernier scale division have same length as 7 main scale division of vernier A and 8 vernier scale division have same length as 3 main scale division for vernier B.
(A) The least count of vernier A is more than the vernier B.
(B) The least count of vernier A is less than the vernier B.
(C) The least count of vernier B is 0.125 mm.
(D) The magnitude of difference in least count of vernier A and vernier B is 0.025 mm.

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. In the given circuit, the value of R so that thermal power generated in R will be maximum is 4.2mΩ. Find the value of m.



12. Velocity of an object in rectilinear motion is given as function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. Its average speed over the time interval from t = 0 second to t = 2 seconds, is $\frac{8K}{27}$ m/s. Find the value of K.

13. An open pipe 68.48 cm long and a closed pipe 58.4 cm long, both having same diameter, are producing their second overtone, and these are in unison. Determine the end correction in centimeter of these pipes.

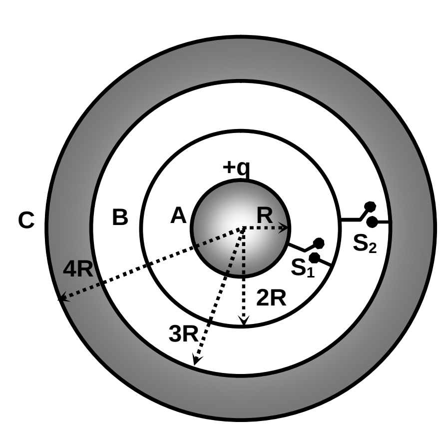
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

Three conducting bodies A(solid sphere), B (spherical shell) and C(hollow sphere with thickness R) are arranged as shown in the figure. A charge q is given to the inner sphere. When S_1, S_2 both are open capacity of the system is C_1 and when S_1 is closed but S_2 is open capacity is C_2, then $(C_2 - C_1) = \frac{X\pi\varepsilon_0 R}{55}$. When S_2 is closed but S_1 is opened capacity is C_3 and When both S_1 and S_2 are closed capacity is C_4 then $(C_4 - C_3) = \frac{Y\pi\varepsilon_0 R}{3}$.



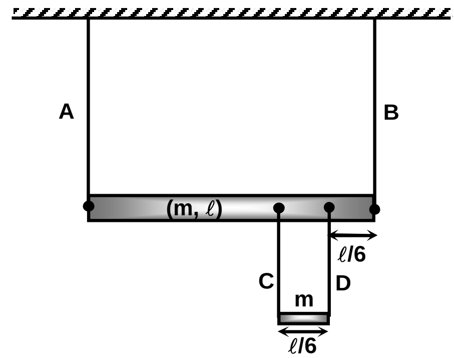
14. The value of X is ………..

15. The value of Y is ………..

Question Stem for Question Nos. 16 and 17

Question Stem

A uniform rod of mass 'm=10kg' and length 'ℓ=20m' is held horizontally by two vertical strings of negligible mass and another rod of mass 'm' is also held by two strings with the rod as shown in the figure. The tension in the string 'A' immediately after the string B and D is cut is X newton and the tension in the string 'A' immediately after the string B and C is cut is Y newton. (Take g = 10 m/s^2)



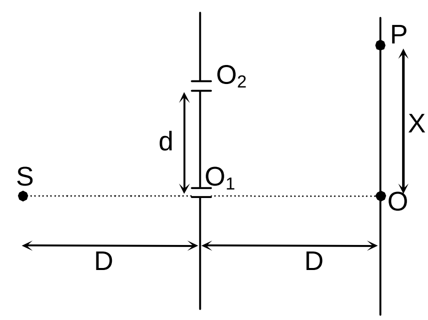
16. The value of X is ………..

17. The value of Y is ………..

Question Stem for Question Nos. 18 and 19

Question Stem

A monochromatic light source is kept at S having wavelength λ in a modified YDSE setup. The minimum value of d so that there is a dark fringe at O is $d_{min} = \sqrt{\dfrac{D\lambda}{M}}$. For the value of d_{min}, the distance of the 2nd nearest bright fringe from O is X(as shown) = $Kd_{min.}$



18. The value of M is………………

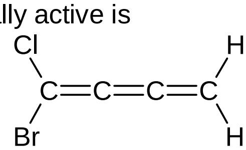
19. The value of K is………………

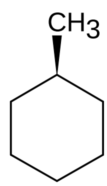
Chemistry PART – II

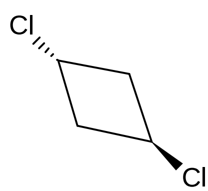
Section – A (Maximum Marks: 12)

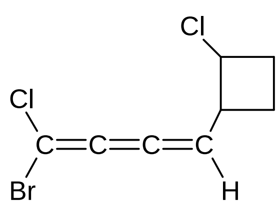
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

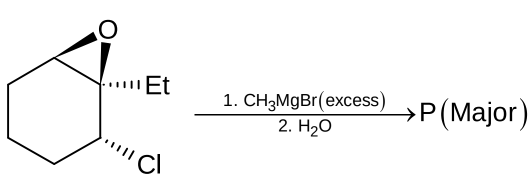
20. Optically active is

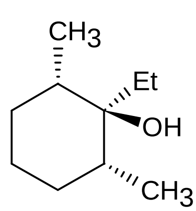
(A) 

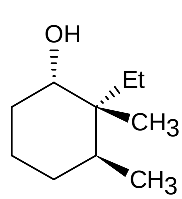
(B) 

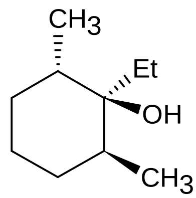
(C) 

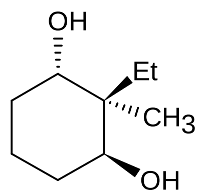
(D) 

21. 

(A) 

(B) 

(C) 

(D) 

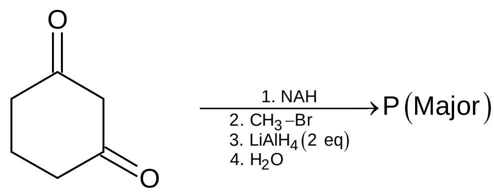
22. 4 mole of a mixture O_2 and O_3 is reacted with excess of acidified solution of KI. The liberated iodine require 1 L of 2 M hypo solution for complete reaction. The mole percent of O_3 in the initial sample is
(A) 25
(B) 30
(C) 75
(D) 50

23. Square pyramidal shape is
(A) XeF_2
(B) ICl_4^-
(C) XeF_5^+
(D) XeF_5^-

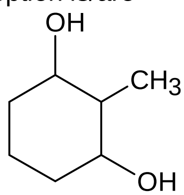
Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

24. High spin complex is/are
(A) $\left[Fe\left(H_2O\right)_6\right]^{2+}$
(B) $\left[FeF_6\right]^{3-}$
(C) $\left[Fe\left(CN\right)_6\right]^{3-}$
(D) $\left[Co\left(H_2O\right)_6\right]^{3+}$

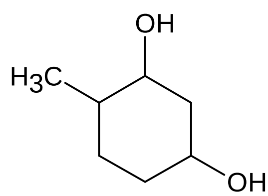
25. 

Choose correct option is/are

(A) P is 

(B) Number stereo isomers in P is 4

(C) Number of stereo isomer in P is 3

(D) P is 

26. The correct statement for $2p_z$ orbital is/are

(A) $\psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2}\left(\frac{Zr}{a}\right)e^{-2r/2a}\cos(\theta)$

(B) $\psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2}\left(2-\frac{Zr}{a}\right)e^{-2r/2a}$

(C) xy plane is nodal plane

(D) $2p_z$ ungerade orbital

27. Which of the following complexes is/are shows the co-ordination isomerism?

(A) $\left[Pt(NH_3)_4\right]\left[PtCl_4\right]$

(B) $\left[CO(en)_3\right]\left[Cr(CN)_6\right]$

(C) $\left[Ag(NH_3)_2\right]\left[AgCl_2\right]$

(D) $\left[CO(NH_3)_6\right]\left[CO(NO_2)_6\right]$

28. Choose homopolymer:

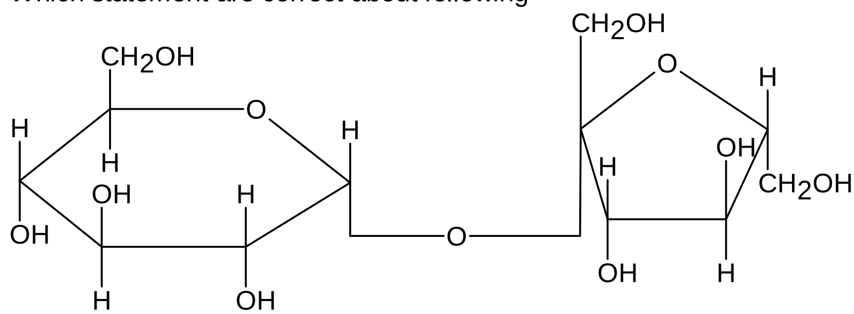
(A) Butadiene-styrene (Buna-S)

(B) Polythene

(C) Polypropene

(D) Nylon 6, 6

29. Which statement are correct about following

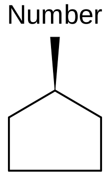


(A) Due to presence of Hemiacetal linkage it is non reducing sugar
(B) It is sucrose
(C) Due to absence of Hemiacetal group it can’t give Tollen’s test
(D) Left side is a $\alpha - \text{D} - \text{Glucose}$ and right side unit is $\beta - \text{D} - \text{fructose}$

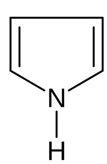
Section – B (Maximum Marks: 12)

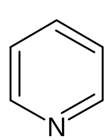
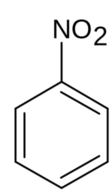
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. Number of monohalogen derivative including stereo isomers for the following is



31. Total number of correct statement among the following is

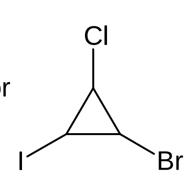
(A)  is aromatic

(B)  is more reactive than  for EAS reaction

(C) In $H_3C-\left(\overset{\overset{O}{\|}}{C}\right)-\dot{C}H_2$ the presence of $-\overset{\overset{O}{\|}}{C}-$ group stabilise the radical

(D) Adiabatic irreversible expansion for ideal gas, $[\Delta S]_{\text{system}} = 0$

(E) In $F-\underset{\underset{F}{|}}{\dot{C}}-F$ the state of hybridization of radical is sp^3

32. Total number of diastereomeric pair for  is x then value of $\frac{x}{6}$ is

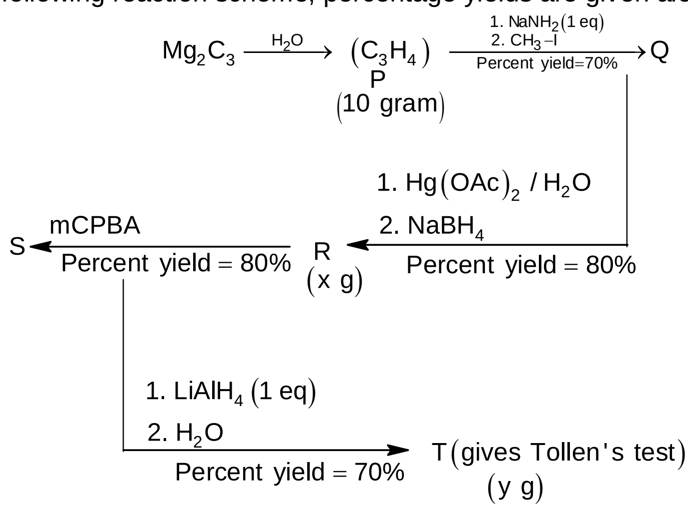
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

For the following reaction scheme, percentage yields are given along the arrow



If mass of R and T are x and y g respectively then

33. The value of x is_______

34. The value of y is_______

Question Stem for Question Nos. 35 and 36

Question Stem

When 2 mole of P(g) is introduced in a closed rigid 1 litre vessel maintained at constant temperature. Following equilibrium are established

$P(g) \rightleftharpoons Q(g) + R(g), \quad K_{C_1} = W$ mole / litre

$R(g) \rightleftharpoons S(g) + Q(g), \quad K_{C_2} = Z$ mole / litre

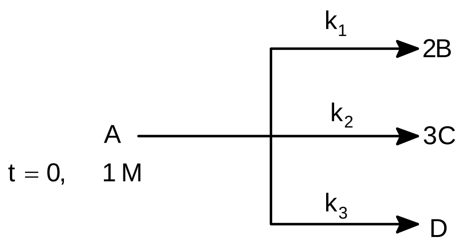
If pressure at equilibrium is twice the initial pressure and $\frac{[R]_{eq}}{[Q]_{eq}} = \frac{1}{5}$

35. The value of W is_________mol/L

36. The value of Z is_________mol/L

Question Stem for Question Nos. 37 and 38

Question Stem



$k_1 = 0.1\ \text{min}^{-1}$

$k_2 = k_3 = 0.2\ \text{min}^{-1}$

If C_t, D_t at $t = 4.606\ \text{min}^{-1}$ are W and Z mol/L

37. The value of W is __________

38. The value of Z is __________

Mathematics PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. $\int_{0}^{2}\left(\sqrt{1+x^3}+\sqrt[3]{x^2+2x}\right)$ is equal to
(A) 4
(B) 5
(C) 6
(D) 7

40. Assume that P = {(x, y) : $x^2 + 2y^2 = 10$} and Q = {(x, y) : y = mx + c}. If $P \cap Q = \phi$ for all $m \in [-1, 1]$, then minimum positive integral value of c is
(A) 3
(B) 4
(C) 5
(D) 10

41. The locus of the centre of the circle which touches the circle $(x-1)^2 + y^2 = 9$ and the line x = 6 is a curve whose directrix is
(A) x = 1
(B) y = 9
(C) y = –4
(D) x = 9

42. If f(x) is a 4 degree polynomial satisfying $f(x)=\frac{x}{x+1}$ for x = 0, 1, 2, 3, 4, then f(6) is
(A) $\frac{12}{7}$
(B) 0
(C) 1
(D) –6

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. The value of definite integral $\int_{-\infty}^{k}\frac{\left(\sin^{-1}e^{x}+\sec^{-1}e^{-x}\right)}{\left(\tan^{-1}e^{k}+\tan^{-1}e^{x}\right)\left(e^{x}+e^{-x}\right)}dx$ (where $k \in R$) is
(A) $\frac{\pi}{2}\ln 2\left(2\tan^{-1}e^{k}\right)$
(B) independent of k
(C) dependent on k
(D) $\frac{\pi}{2}\ln 2$

44. Let g(x) be a cubic polynomial with leading coefficient unity such that the roots of g(x) = 0 are the squares of the roots of $x^3 + x + 1 = 0$, then
(A) g(3) is a prime number
(B) last two digit of $(g(1))^{50}$ are 49
(C) when $(g(2))^{2022}$ is divided by 5, remainder is 2
(D) g(x) = 0 has all 3 roots real

45. Let P and Q be two square matrix of order 3, then which of the following statement is/are always CORRECT?
(A) PQP^T is symmetric matrix
(B) PQ – QP is skew symmetric matrix
(C) If $Q = |P|P^{-1}$, $|P| \neq 0$, then $adj(P^T) - Q$ is skew symmetric matrix
(D) If $Q + P^T = 0$ and P is skew symmetric matrix and Q^{15} is also skew symmetric matrix

46. A tangent L_1 is drawn to the curve $x^2 - 4y^2 = 16$ at point A in first quadrant whose abscissa is 5. Another tangent L_2 parallel to L_1 meets the curve at B. L_3 and L_4 are normal to the curve at A and B lines L_1, L_2, L_3, L_4 forms a rectangle, then
(A) equation of normal at B is 12x + 10y + 75 = 0
(B) area of rectangle $L_1 L_2 L_3 L_4$ is $\frac{2400}{61}$
(C) radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$
(D) radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$

47. The integral point satisfying $|z - 1| + |z + 1| \leq 6$ in the argand plane for which $||z + \omega^2| - |z + \omega||$ is minimum (where ω is complex root of unity) is/are
(A) (2, 0)
(B) (1, 2)
(C) (–3, 0)
(D) (0, 2)

48. Let for $x \in [1, \infty)$, $\int_1^x f(x)dx = \left(\frac{x^2+1}{2x}\right)\left(f(x) - x + \frac{2x}{x^2+1}\right) = \ln 2$, then for $x \in [1, \infty)$ which of the following is are TRUE?
(A) f(1) = ln 2
(B) Range of $g(x) = \frac{xf'(x) - f(x)}{x}$ is [0, 2]
(C) $\lim_{x\to 3}\left[\frac{f(x)}{x}\right] = 2$
(D) $h(x) = \frac{f(x)}{x}$ is an odd function
(where [.] denotes greatest integer function)

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. If P is the number of ways in which a person can walk up a stairway which has 11 steps if he can take 1 or 2 steps up the stairs at a time, then $\frac{P}{16}$ is equal to

50. In a triangle ABC, side AC = 4 units and sin A sin B + sin B sin C + sin C sin A = $\frac{9}{4}$. If A is the area or the triangle ABC, then $\frac{A}{\sqrt{3}}$ is equal to

51. If p is the least positive integer which satisfies the equation $|x - 2| + |x^2 - 9x + 20| = |x^2 - 8x + 18|$ and q is the minimum value of $\sin^2 x + \sin x + 1$ and α is the only positive root of $\left((2021)^{2022} - 1\right)x^2 + \left(p - (2021)^{2022}\right)x = 1$, then the value of $\left(\alpha^{2022} - 1\right)p^2 + \left(p - 1 - \alpha^{2017}\right)pq + 8q$ is

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $A_n = [A_{ij}]$ be a $n \times n$ matrix such that $A_{ij} = \begin{cases} k & ; \quad i = j \\ 1 & ; \quad |i - j| = 1 \\ 0 & ; \quad \text{otherwise} \end{cases}$. Let B_n denote the determinant of matrix A_n

52. If k = 2, then B_{2022} is equal to

53. If k = 1, then $\sum_{p=1}^{2022} |B_p|$ is equal to (where |.| denotes absolute value function

Question Stem for Question Nos. 54 and 55

Question Stem

Let $A_1, A_2, A_3, \ldots A_{12}$ be a regular polygon of 12 sides whose centre is origin. Let the complex numbers representing vertices $A_1, A_2, A_3, \ldots A_n$ be $z_1, z_2, z_3, \ldots z_{12}$ respectively. Let $OA_1 = OA_2 = OA_3 = \ldots OA_{12} = 1$ (where O is origin)

54. The value of $|A_1 A_2| \times |A_1 A_3| \times |A_1 A_4| \times \ldots |A_1 A_{12}|$ is equal to

55. The value of $|A_1 A_2|^2 + |A_1 A_3|^2 + \ldots |A_1 A_n|^2$ is

Question Stem for Question Nos. 56 and 57

Question Stem

Suppose 12 students of a class are asked a question successively in a random order and exactly 4 of them known the answer. Assume that none of them are aware of the answers given by other students, answer the following

56. If the probability that 6^{th} student to be asked the question is the first one to know the answer is k, then 99k is equal to

57. If the probability that 1^{st} and 12^{th} student knows the answer is k, then 99k is equal to